

Chapter 15

Localized currents and magnetic dipoles

15.1 Multipole expansion for localized currents

For a localized current distribution observed from far away, we expand

$$\frac{1}{|\vec{x} - \vec{x}'|} = \frac{1}{x} + \frac{\vec{x} \cdot \vec{x}'}{x^3} + \cdots \quad , \quad x \equiv |\vec{x}| \quad (15.1)$$

so that the vector potential

$$\vec{A}(\vec{x}) = \frac{1}{c} \int \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} dV' \quad (15.2)$$

becomes

$$\vec{A} = \frac{1}{c} \int \vec{J}(\vec{x}') dV' + \frac{1}{c} \int \vec{J}(\vec{x}') \frac{\vec{x} \cdot \vec{x}'}{x^3} dV' \quad (15.3)$$

The first term vanishes. Proof: using $\vec{\nabla} \cdot \vec{J} = 0$,

$$\begin{aligned} 0 = \int x \vec{\nabla} \cdot \vec{J} dV &= \int \vec{\nabla} \cdot (x \vec{J}) dV - \int J^x dV \\ &= \oint x \vec{J} \cdot d\vec{a} - \int J^x dV \end{aligned} \quad (15.4)$$

and the surface term vanishes for localized currents – hence $\int J^x dV = 0$, and similarly for the other components. We are left with

$$\vec{A} = \frac{1}{c x^3} \int \vec{J}(\vec{x}') (\vec{x} \cdot \vec{x}') dV' \quad (15.5)$$

Using the separation identity (15.13) (proved below), this becomes

$$\vec{A} = -\frac{\vec{x}}{2 c x^3} \times \int \vec{x}' \times \vec{J}(\vec{x}') dV' \quad (15.6)$$

15.2 The magnetic dipole

Define the *magnetic dipole moment*

$$\vec{m} \equiv \frac{1}{2 c} \int \vec{x}' \times \vec{J}(\vec{x}') dV' \quad (15.7)$$

so that (15.6) reads

$$\vec{A} = \frac{\vec{m} \times \vec{x}}{x^3} \quad (15.8)$$

The magnetic field is then

$$\vec{B} = \vec{\nabla} \times \vec{A} = -\frac{\vec{m}}{x^3} + \frac{3 \vec{x} (\vec{m} \cdot \vec{x})}{x^5} [?] \quad (15.9)$$

For a planar loop carrying current I ,

$$\vec{m} = \frac{1}{2c} \int \vec{x}' \times \vec{J}(\vec{x}') dV' \rightarrow \frac{1}{2c} \oint \vec{x}' \times I d\vec{s}' = \frac{I}{2c} \oint \vec{x}' \times d\vec{x}' = \frac{I \vec{a}}{c} \quad (15.10)$$

where $\vec{a}$ is the area vector of the loop, oriented by the right-hand rule.

15.3 Magnetic force

The magnetic force on a point charge generalizes to line, surface, and volume currents as

$$\vec{F} = \frac{q \vec{v}}{c} \times \vec{B} \rightarrow \frac{1}{c} \int I d\vec{s} \times \vec{B} \rightarrow \frac{1}{c} \int \vec{K} \times \vec{B} da \rightarrow \frac{1}{c} \int \vec{J} \times \vec{B} dV \quad (15.11)$$

15.4 Identities

Using $\vec{\nabla}' \cdot \vec{J}(\vec{x}') = 0$, we have the identities

$$\int \vec{J}(\vec{x}') dV' = 0 \quad (15.12)$$

$$\int (\vec{x} \cdot \vec{x}') \vec{J}(\vec{x}') dV' = -\frac{1}{2} \vec{x} \times \int \vec{x}' \times \vec{J}(\vec{x}') dV' \quad (15.13)$$

$$\int (x'^i J^j + x'^j J^i) dV' = 0 \quad (15.14)$$

These are proved using the identity

$$\int \left(f \vec{J} \cdot \vec{\nabla}' g + g \vec{J} \cdot \vec{\nabla}' f \right) dV' = 0 \quad \text{for all } f, g \quad (15.15)$$

with $f = 1$, $g = x'^i$, or $f = x'^i$, $g = x'^j$.

15.5 Force on localized currents

Consider a localized current distribution in an external magnetic field that varies slowly over the currents. Expand the field about the origin

$$\vec{B}(\vec{x}') = \vec{B}(0) + \left(\vec{x}' \cdot \vec{\nabla}' \right) \vec{B} \Big|_0 + \cdots \quad (15.16)$$

Then the force becomes

$$\vec{F} = -\frac{1}{c} \vec{B}(0) \times \int \vec{J}(\vec{x}') dV' - \frac{1}{c} \int \left(\vec{x}' \cdot \vec{\nabla}' \right) \vec{B}(0) \times \vec{J}(\vec{x}') dV' \quad (15.17)$$

The first term vanishes by (15.12); in the second, the derivative acts on $\vec{B}$ first and is then evaluated at the origin, so there is no $\vec{x}'$ dependence left in $\vec{B}$. But the external field satisfies $\vec{\nabla} \times \vec{B} = 0$ at the currents, so

$$\vec{x}' \times \left(\vec{\nabla} \times \vec{B} \right) = 0 = \vec{\nabla} \left(\vec{x}' \cdot \vec{B} \right) - \left(\vec{x}' \cdot \vec{\nabla} \right) \vec{B} \quad (15.18)$$

Hence

$$\begin{aligned} \vec{F} &= \frac{1}{c} \int \vec{J}(\vec{x}') \times \vec{\nabla} \left(\vec{x}' \cdot \vec{B} \right) dV' = -\frac{1}{c} \vec{\nabla} \times \int \left(\vec{x}' \cdot \vec{B} \right) \vec{J}(\vec{x}') dV' \\ &= \vec{\nabla} \times \left(\vec{B} \times \frac{1}{2c} \int \vec{x}' \times \vec{J}(\vec{x}') dV' \right) = \vec{\nabla} \times \left(\vec{B} \times \vec{m} \right) \end{aligned} \quad (15.19)$$

using (15.13) in the last line, with $\vec{m}$ the magnetic dipole moment (15.7). Expanding the double cross product,

$$\vec{F} = (\vec{m} \cdot \vec{\nabla}) \vec{B} - (\vec{\nabla} \cdot \vec{B}) \vec{m} = (\vec{m} \cdot \vec{\nabla}) \vec{B} = \vec{\nabla} (\vec{m} \cdot \vec{B}) \quad (15.20)$$

where the middle term drops since $\vec{\nabla} \cdot \vec{B} = 0$, and the last step uses $\vec{\nabla} \times \vec{B} = 0$ again. The corresponding energy is

$$U = -\vec{m} \cdot \vec{B} \quad (15.21)$$

15.6 Torque

The torque on the current distribution is

$$\vec{N} = \int \vec{x}' \times \frac{\vec{J}(\vec{x}') \times \vec{B}}{c} dV' = \frac{1}{c} \int \left[(\vec{x}' \cdot \vec{B}) \vec{J}(\vec{x}') - (\vec{x}' \cdot \vec{J}(\vec{x}')) \vec{B} \right] dV' \quad (15.22)$$

The second term vanishes since

$$\int \vec{x}' \cdot \vec{J}(\vec{x}') dV' = \frac{1}{2} \int \vec{\nabla}' \cdot (x'^2 \vec{J}(\vec{x}')) dV' = \frac{1}{2} \oint x'^2 \vec{J} \cdot d\vec{a} = 0 \quad (15.23)$$

using $\vec{\nabla}' \cdot \vec{J} = 0$ and $\vec{\nabla}' \cdot (x'^2 \vec{J}) = 2(\vec{x}' \cdot \vec{J}) + x'^2 \vec{\nabla}' \cdot \vec{J}$. For the first term, the separation identity (15.13) gives

$$2 \int \vec{J}(\vec{x}') (\vec{B} \cdot \vec{x}') dV' = -\vec{B} \times \int \vec{x}' \times \vec{J}(\vec{x}') dV' \quad (15.24)$$

so that

$$\vec{N} = -\vec{B} \times \int \frac{\vec{x}' \times \vec{J}(\vec{x}')}{2c} dV' = \vec{m} \times \vec{B} \quad (15.25)$$

15.7 Two-dipole interaction energy

As a remark, the interaction energy of two magnetic dipoles separated by $\vec{x}$ is

$$U = - \left(\frac{3(\vec{m}_1 \cdot \vec{x})(\vec{m}_2 \cdot \vec{x})}{x^5} - \frac{\vec{m}_1 \cdot \vec{m}_2}{x^3} [?] \right) \quad (15.26)$$